



Handout 1.13: Nacho Label Math

Lesson: 1.13 Lab 2, Day 1 (Step 3) and the label discussion during the bake on Day 2 **Print:** One per student. The Label Reader leads; everyone computes. A calculator is allowed.



Name: _____ Date: _____ Period: _____ Station: _____

The question: How much sodium is in ONE person's plate of the nachos we are making? Is this a snack to eat often or sometimes?

Directions: Use the four packages on your counter (chip bag, cheese bag, bean can, corn can). Find the serving size and the sodium per serving on each Nutrition Facts label. Then multiply by the number of servings OUR STATION puts in the pan (that number is already in the table; it is not the same as servings per container). Add. Divide by 7. Add the salsa bar. Compare to a whole day.

The numbers printed below are typical store brand numbers. **Use the numbers on the real packages if they are different, and cross these out.**

Part 1: Four labels, four lines

Ingredient	Serving size on the label	Sodium per serving (mg)	Servings our station uses	Sodium our station puts in the pan (mg)
Tortilla chips (1 bag, 13 oz)	1 oz (about 7 chips)	115	13	13 x _____ = _____
Shredded cheese (1 cup)	1/4 cup (28 g)	180	4	4 x _____ = _____
Black beans (1/2 can)	1/2 cup (130 g)	400	1.75 (grade 6: use 2)	1.75 x _____ = _____
Corn (1/3 can)	1/2 cup (125 g)	200	1	1 x _____ = _____

The full labels, in case a package is missing:

Tortilla chips, restaurant style, 13 oz bag. Serving size 1 oz (28 g, about 7 chips). Servings per container about 13. Calories 140. Total fat 7 g (9%). Saturated fat 1 g (5%). Sodium 115 mg (5%). Total carbohydrate 19 g (7%). Dietary fiber 1 g (4%). Total sugars 0 g. Added sugars 0 g. Protein 2 g. Calcium 20 mg (2%). Iron 0.4 mg (2%). Ingredients: corn, vegetable oil, salt.

Shredded Mexican blend cheese, 8 oz bag. Serving size 1/4 cup (28 g). Servings per container 8. Calories 100. Total fat 8 g (10%). Saturated fat 5 g (25%). Cholesterol 25 mg (8%). Sodium 180 mg (8%). Total carbohydrate less than 1 g. Protein 6 g. Calcium 200 mg (15%). Ingredients: cheddar, Monterey Jack, mozzarella, and asadero cheeses (pasteurized milk, cheese culture, salt, enzymes), potato starch and powdered cellulose (to prevent caking), natamycin.

Black beans, 15.5 oz can. Serving size 1/2 cup (130 g). Servings per container about 3.5. Calories 110. Total fat 0.5 g (1%). Sodium 400 mg (17%). Total carbohydrate 20 g (7%). Dietary fiber 7 g (25%). Total sugars 1 g. Added sugars 0 g. Protein 7 g. Calcium 40 mg (4%). Iron 2 mg (10%). Potassium 350 mg (8%). Ingredients: prepared black beans, water, salt, calcium chloride.

Corn, whole kernel, 15.25 oz can. Serving size 1/2 cup (125 g). Servings per container about 3.5. Calories 60. Total fat 1 g (1%). Sodium 200 mg (9%). Total carbohydrate 13 g (5%). Dietary fiber 2 g (7%). Total sugars 5 g. Added sugars 0 g. Protein 2 g. Ingredients: corn, water, salt.

Part 2: The pan and the plate

Sodium in the WHOLE PAN: ____ + ____ + ____ + ____ = ____ mg

One person's share (divide by 7): ____ divided by 7 = ____ mg

Add the salsa bar (one serving each): salsa 2 tablespoons = 220 mg; sour cream 2 tablespoons = 15 mg.

One person's PLATE: ____ + 220 + 15 = ____ mg

Part 3: Compare to a whole day

The daily value for sodium is 2,300 mg. That is the "100%" on every label.

My plate divided by 2,300, times 100 = ____% of a whole day's sodium, from one snack.

(Grade 6: the percent is about 30%. Check that your plate number is close to 680 mg.)

Part 4: Judge it

Circle one: This is an **OFTEN** snack / a **SOMETIMES** snack.

Why? Use the number. _____



The beans surprise. The black beans are the healthiest thing on the pan (7 g fiber, 7 g protein, iron) and one of the saltiest. Rinsing canned beans washes away about 40% of the sodium. If the Prep Cook rinses, about how much sodium leaves the pan? 40% of ____ (the beans line) = about ____ mg.

One swap. If you could change ONE ingredient to cut the sodium most, which one and what would you swap it for? (Ideas: low sodium beans, about 130 mg per serving; no salt added corn, about 10 mg; a reduced sodium chip, about 65 mg; less cheese.)

Swap: _____ New per-person plate: about ____ mg

Grade 8 stretch: do the same table for calories and for saturated fat. Whole pan calories: _____. Per person: _____. Saturated fat per person: _____ g, which is _____% of the daily value (20 g).

Teacher key

Part 1, with the printed numbers. Chips $13 \times 115 = 1,495$ mg. Cheese $4 \times 180 = 720$ mg. Beans $1.75 \times 400 = 700$ mg (grade 6 with 2 servings: 800 mg). Corn $1 \times 200 = 200$ mg.

Part 2. Whole pan: $1,495 + 720 + 700 + 200 = 3,115$ mg (grade 6: 3,215 mg). Per person: $3,115$ divided by $7 = 445$ mg (grade 6: 459 mg). Plate with salsa bar: $445 + 220 + 15 = 680$ mg (grade 6: 694 mg). Accept anything from 600 to 750 mg if the real packages differ a little.

Part 3. 680 divided by $2,300 = 0.296$, about 30%.

Part 4. Sometimes is the defensible answer: one snack is almost a third of a day's sodium, and it has 13 servings of chips in the pan. A student who argues "often" with a plan to rinse the beans, use low sodium beans, and skip the salsa can get there under 400 mg; accept it if the math is shown. The beans surprise: 40% of 700 = about 280 mg leaves the pan, about 40 mg per person. The swap with the biggest effect is the chips (13 servings; a reduced sodium chip at 65 mg cuts the pan by 650 mg, about 93 mg per person), then salsa (skip it: minus 220 per plate, the biggest single per-plate cut). Expect students to pick the beans first because 400 is the biggest number on one line; that is the discussion: per serving versus in the pan.

Grade 8 stretch. Calories: chips $13 \times 140 = 1,820$; cheese $4 \times 100 = 400$; beans $1.75 \times 110 = 193$; corn 60. Pan 2,473; per person 353, plus salsa 10 and sour cream 60 = 423 calories per plate. Saturated fat: chips $13 \times 1 = 13$ g; cheese $4 \times 5 = 20$ g; beans 0; corn 0; pan 33 g; per person 4.7 g, plus sour cream 3.5 g = 8.2 g, about 41% of the daily value of 20 g. That number usually surprises them more than the sodium.

Rubric row 3 (Lab 2 rubric). Meets: table complete, per-person plate number within 600 to 750 mg, judgment with the number. Exceeds: also a swap with the new number computed.